Jonathan Peris Beltrán

Game Programmer — C++, graphics, engine systems and gameplay AI

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Profile

Third-year game programming student at ESAT (Pearson HND, graduating 2027) with a previous Higher Technician degree in software development and a year of professional experience. I work mostly in C++, from my own data structures and math libraries to procedural terrain in OpenGL and gameplay systems in Unreal Engine 5. Aiming to specialise in gameplay AI.

Projects

Procedural Open-World Renderer — 2025/26

C++, OpenGL, GLSL · Team of 3 · Video and write-up on the portfolio

- Built the procedural terrain: heightmaps from fractal Simplex noise, chunk meshes and the terrain shader (height-based biomes, Blinn-Phong lighting).
- Built the 4×4 chunk grid that streams endless terrain around the player, rebuilding only the edge left behind, with seamless borders in geometry and lighting.
- Runs at about 60 FPS on an RTX 2070 Super, on a custom GPU layer over the school's EDK framework and my own math library.

Data Structures & Algorithms Library — 2025/26

C, ported to C++ · Team of 2

- Implemented the core ADTs (memory node, vector, singly and doubly linked lists, stack, queue) and Quicksort.
- Benchmarked vector vs. lists over 10,000+ runs per operation; found chained vector concatenation ~3× slower because of repeated reallocation.

Math Library — 2025/26

C++ · Solo

- Vec2/3/4 and Mat2/3/4 with inline operations for 2D/3D transforms, covered by a unit-test project; used by the renderer above.

Current work

- **Final-year game (UE5, C++)** — top-down puzzle game in a multidisciplinary team, versioned with Perforce. I own the aiming and weapon components, built for designers to tune, and will take on enemy AI.
- **Custom graphics engine (C++)** — built in a team of two; I own the first render path and the input system.
- **Reinforcement-learning home-automation agent** — solo AI project that learns household routines and balances comfort against energy use.

Experience

Application Developer · SofTPV — Apr 2023 – Apr 2024

- Developed a management application for hospitality businesses in Java and .NET.

IT & Networks Technician · Tecnohostel — Apr 2021 – Apr 2023

- Installed and configured computer systems and networks for hospitality clients.

Education

HND in Computing — Video Game Programming (Pearson) · ESAT, Valencia — 2024 – 2027

Graphics, low-level (ARM / GBA), algorithms & data structures, game engines, AI for games, networking, technical art.

Higher Technician in Multiplatform Application Development (CFGS DAM) · Aula Campus — 2020 – 2022

Course project: Android calorie-tracking app in Java (BMR from height and weight, meal log, deficit or surplus).

Technician in Microcomputer Systems and Networks (CFGM SMR) · Academia Almi — 2018 – 2020

Skills

Languages	C++ (main), C, C# / .NET, Java, Python (basic), GLSL, ARM assembly, SQL
Engines & APIs	Unreal Engine 5 (C++ and Blueprints, materials, Niagara, IK retargeting), OpenGL, Unity
Core	Data structures & algorithms, linear algebra for 3D, memory management, debugging, unit testing
Tools	Visual Studio, Git, Perforce, Premake, Doxygen
Spoken	Spanish (native), Catalan (native), English (B2, CEFR)